

Business Management Systems

Networks

Riaz Esmailzadeh

We respectfully acknowledge the Kaurna, Boandik, and Barngarla First Nations Peoples and their Elders past and present, who are the Traditional Owners of the lands that are home to our campuses across Adelaide and South Australia.

Outline

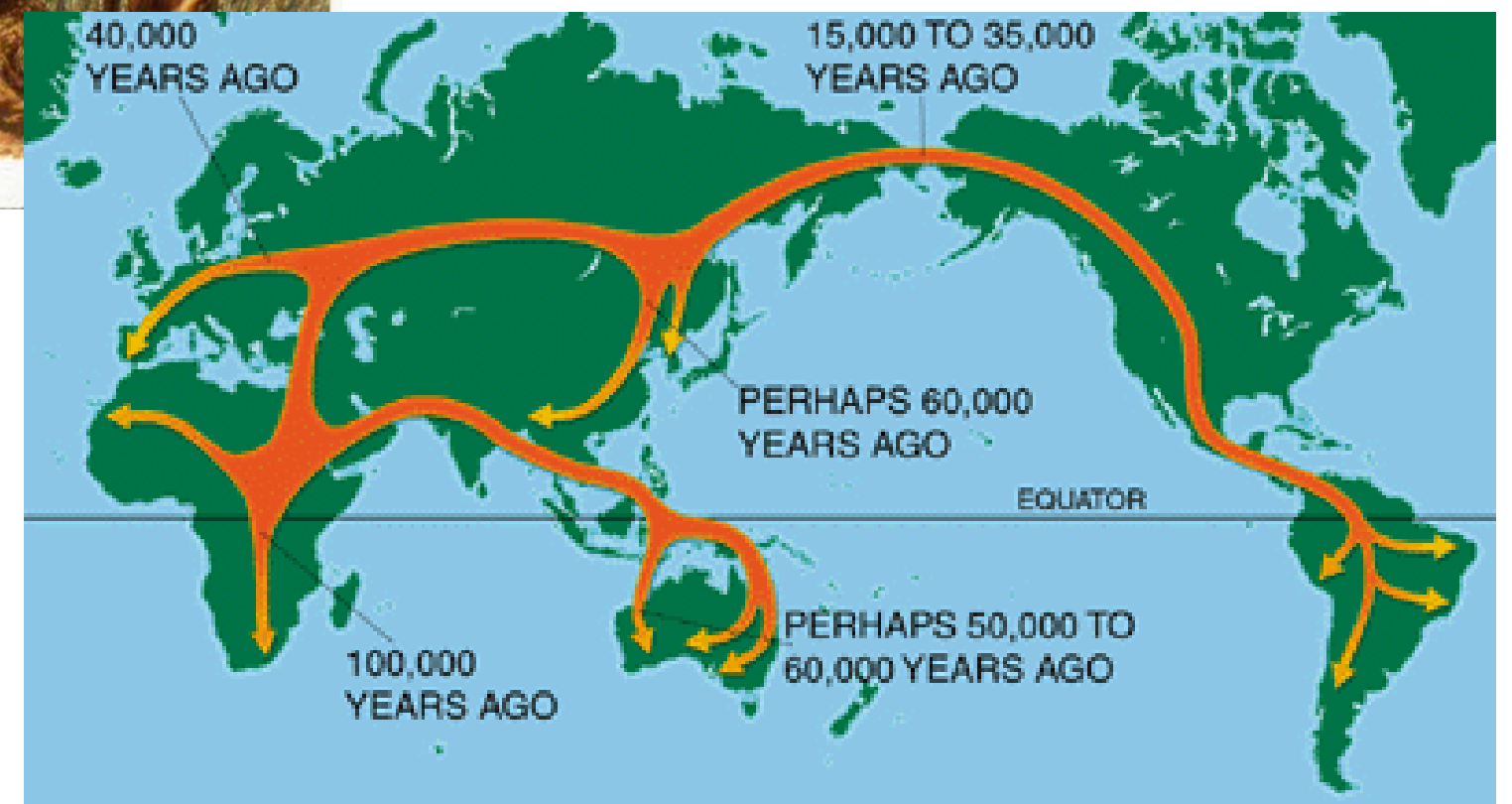
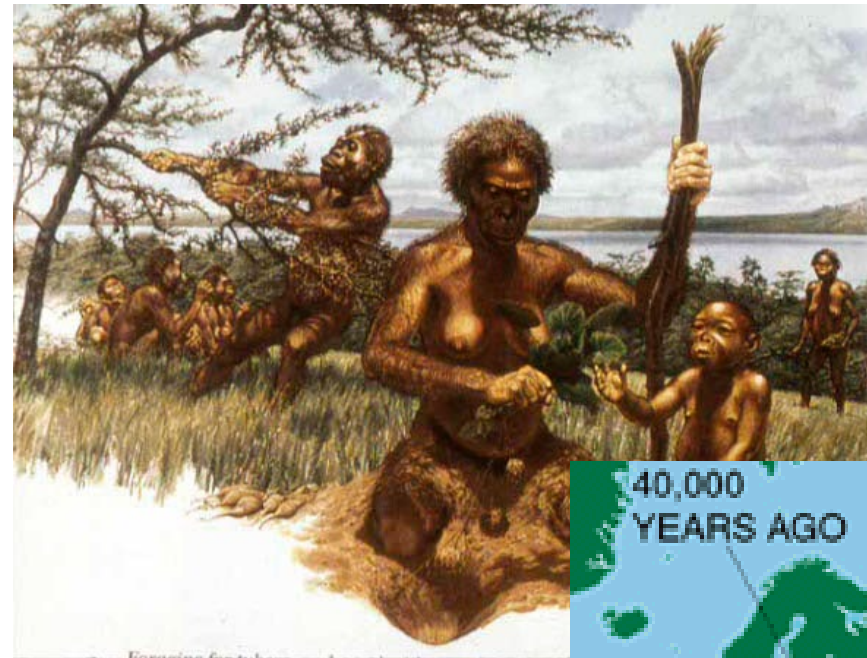
- How do we work together
- Structure and networks
- Some basic network science

Outline

- How do we work together
- Structure and networks
- Some basic network science

Hunters and Gatherers

- We started off around a 100,000 years ago in Africa
- Foraged for food through hunting and gathering
- And then spread around the world



Villages

- Where certain wild seeds could be found, food production through farming, and raising domesticated animals led to higher concentrations of population.
- These happened with the emergence of villages, and creation of hierarchical societies



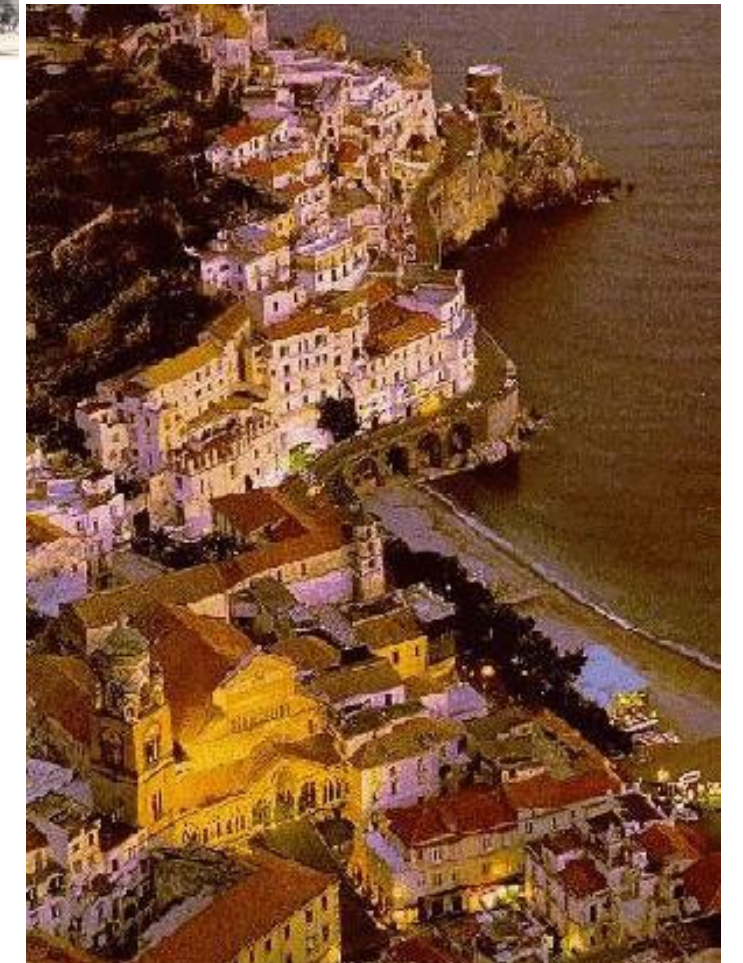
Growth of Civilizations

- Initial civilizations grew in areas with large populations
- Here is where modern Ethical systems were articulated, and religions systems emerged



City-States

- Further growth in population, caused by better food production, led to even more sophisticated society structures
- Rules of behaviour between society members, and between the ruler and the ruled evolved



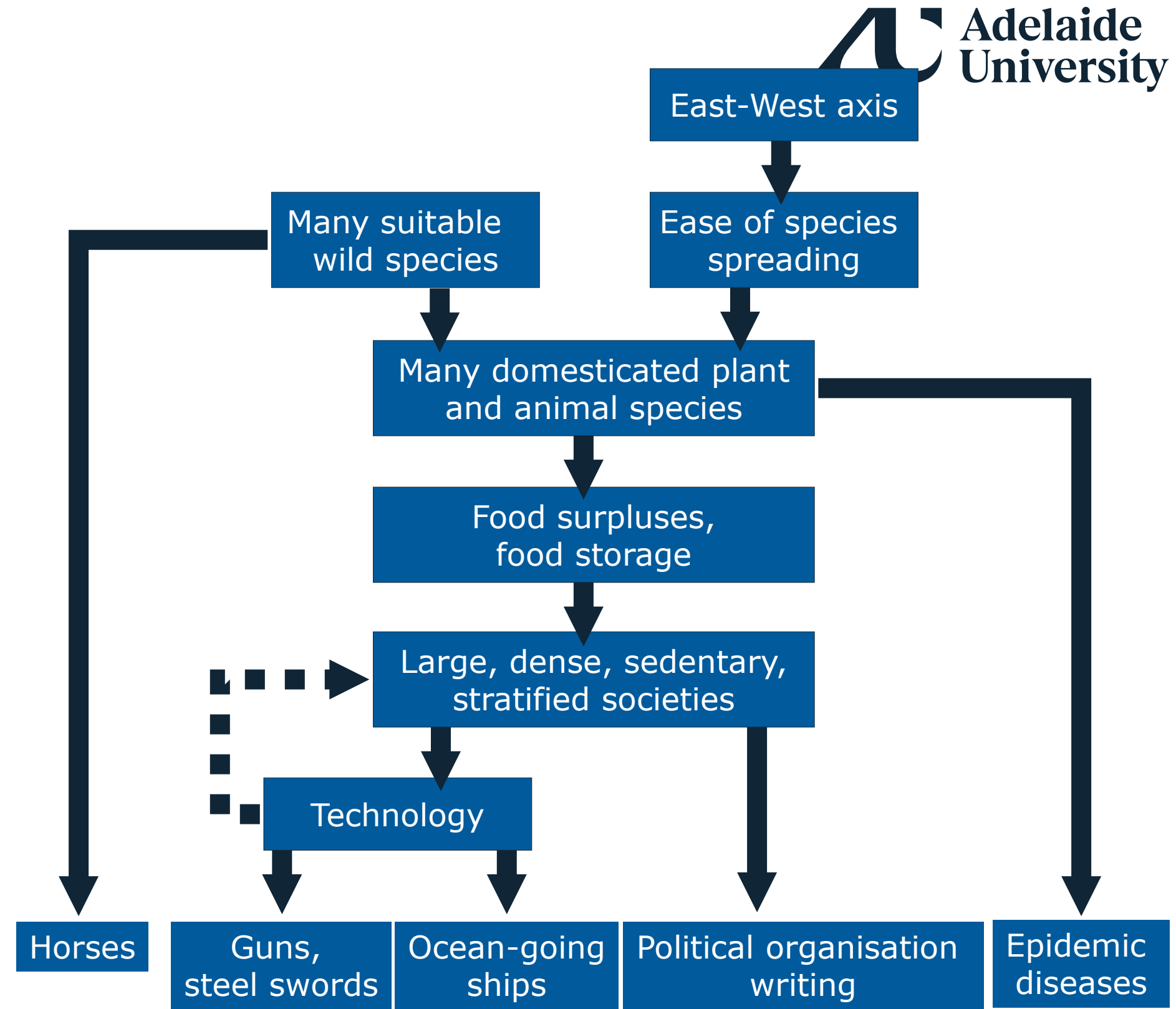
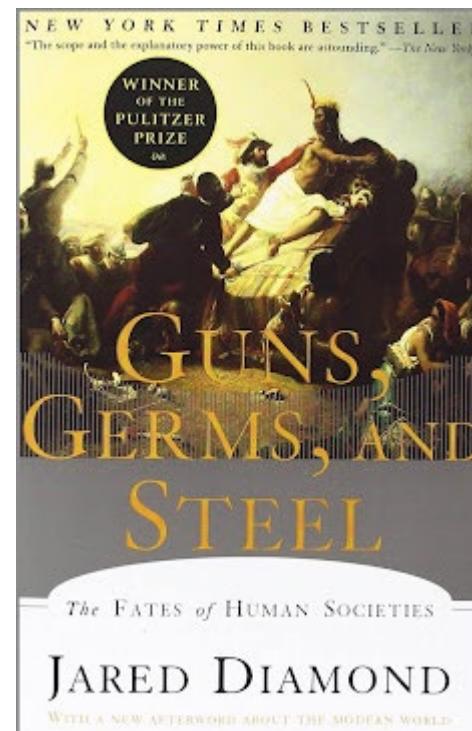
Nation-States

- Nation states are a relatively new social development in most parts of the world
- Diamond's question:
 - Why did this process happen in some areas faster than others?



Diamond's Answer

- There are ultimate factors and because of that proximate factors



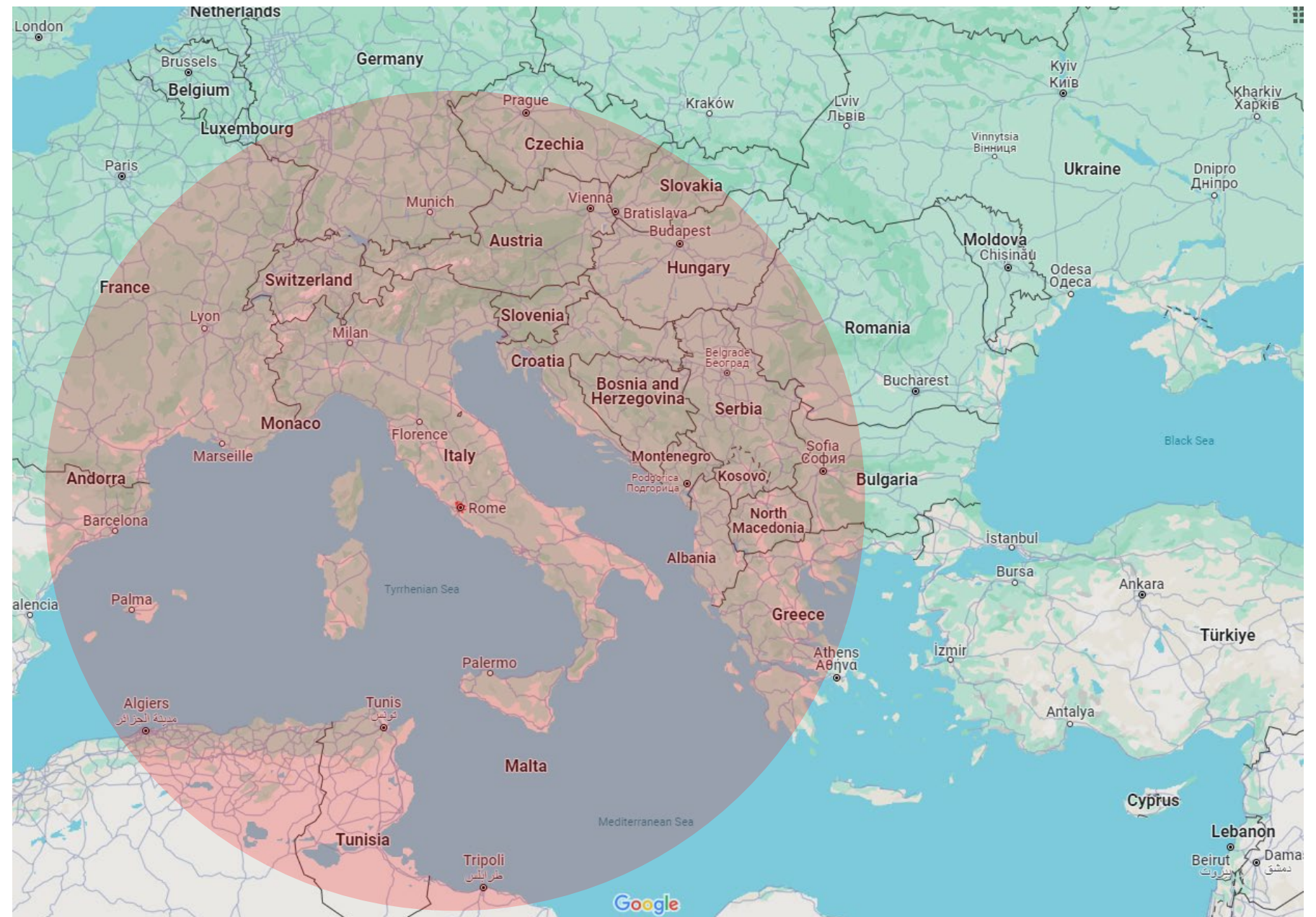
Source: Diamond, "Guns, Germs, and Steel"

The Colosseum



The Supply of Material

- It is estimated that the work involved some 50,000 workers
- All of power was supplied through human and animal labour
- Food and material was brought over from a radius of 1000 kilometers



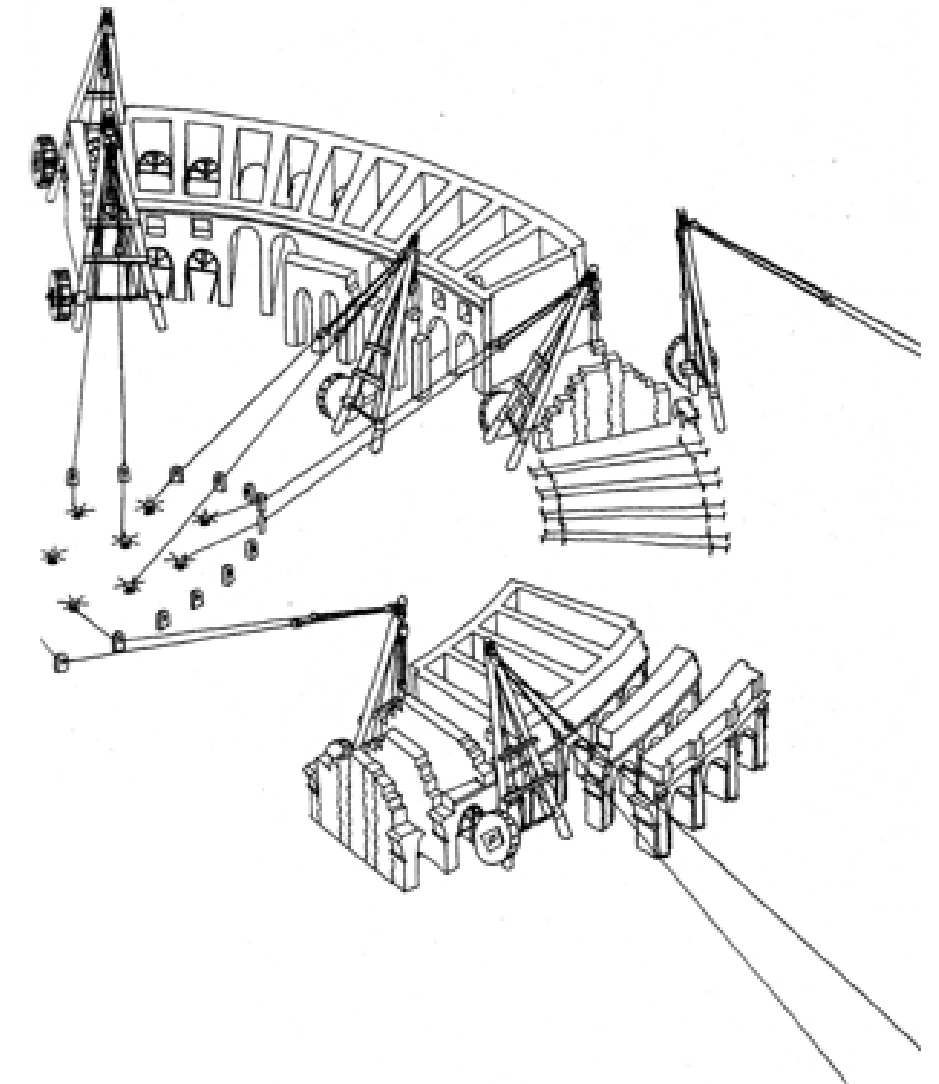
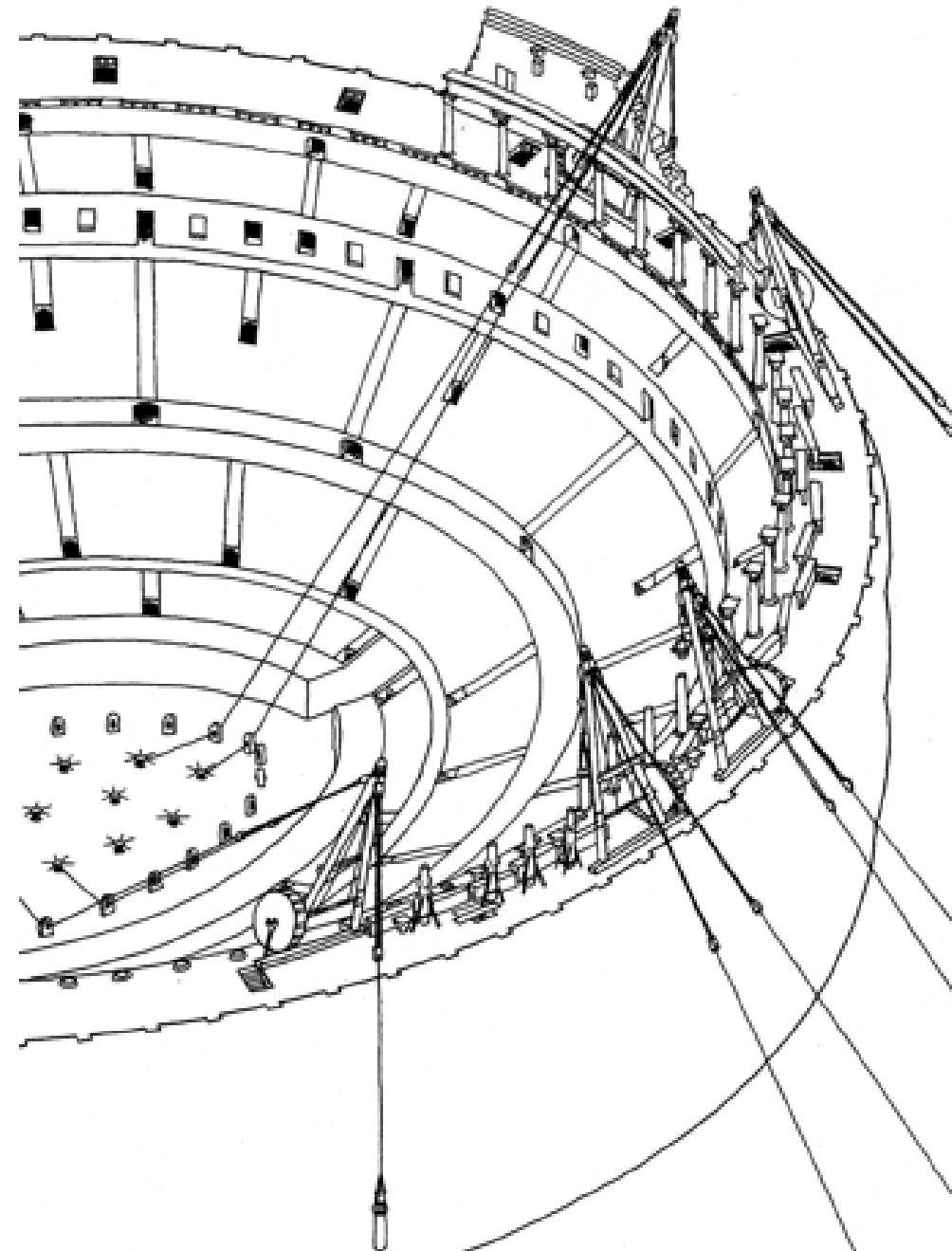
Stones from Far Away...

- This is the Hajar al-Hibla, from a Roman quarry
- Note it has been carved and ready to be transported
 - Except that the Romans run out of money...



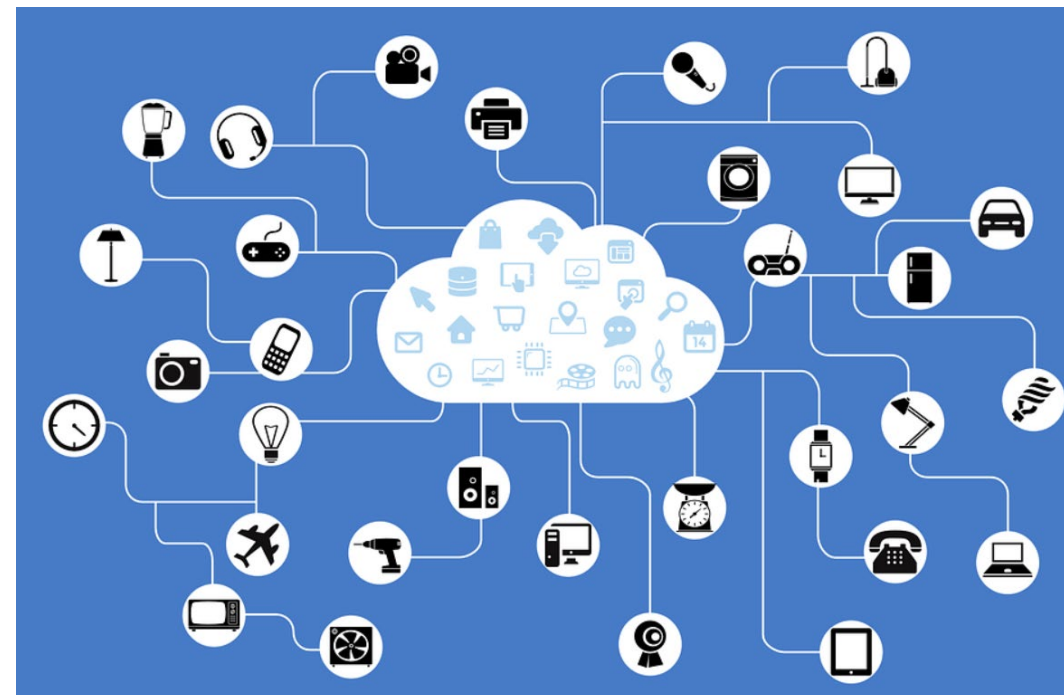
Great Engineering Projects...

- The amount of infrastructure that needed to be constructed to build the Colosseum, without the help of all modern machinery that we have is mind-boggling...



Same Goes for





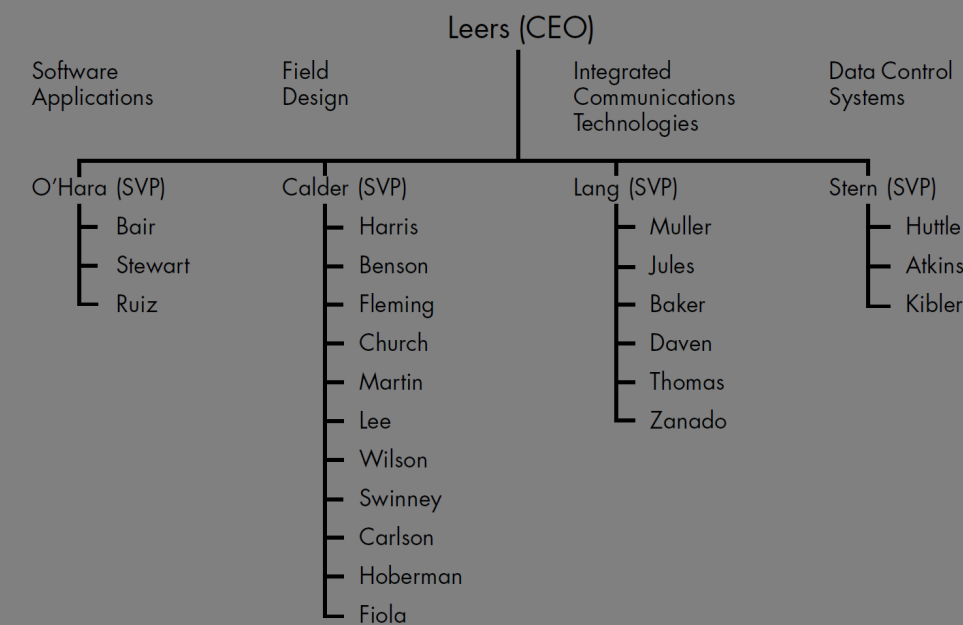
Mobilisation of People

- All of these structures require mobilisation of large groups of people
 - Using hierarchical management structures
- And much engineering knowledge
- And much investment of resources
- All towards a structures that provides some value to the society
 - (Although we can debate how valuable that is...)

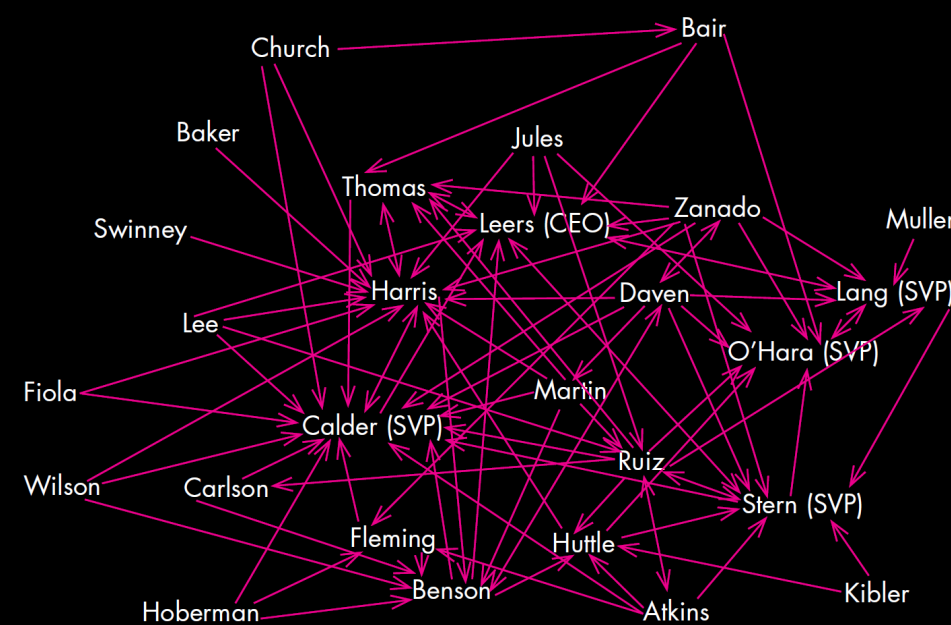
Organization Structures

- Formal and informal (social) networks are present in an organizations.
- Through surveys one can establish advice and trust networks within the organizations

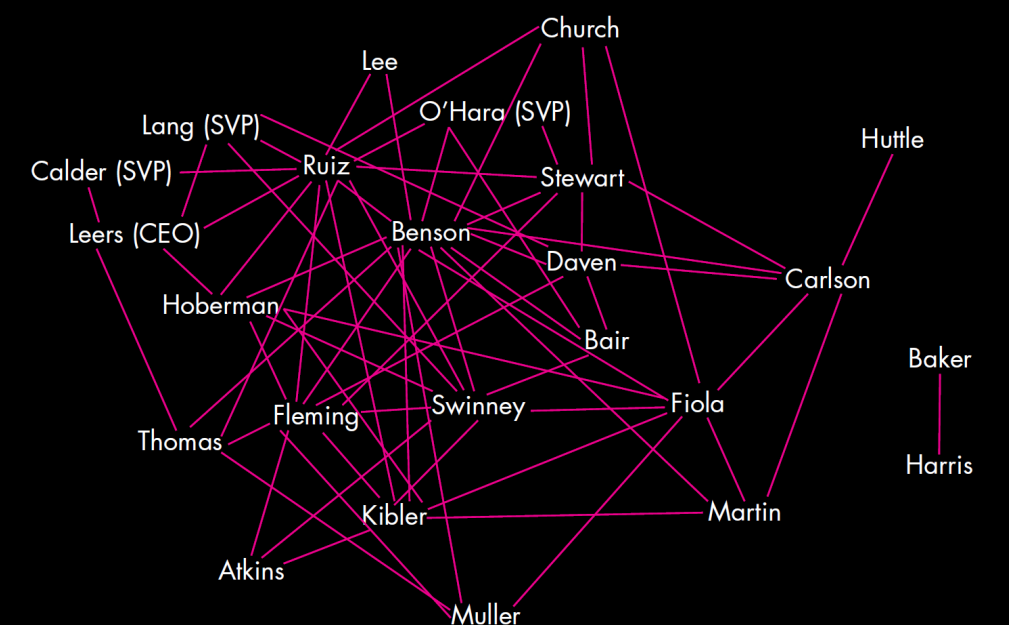
The Formal Chart Shows Who's on Top



The Advice Network Reveals the Experts



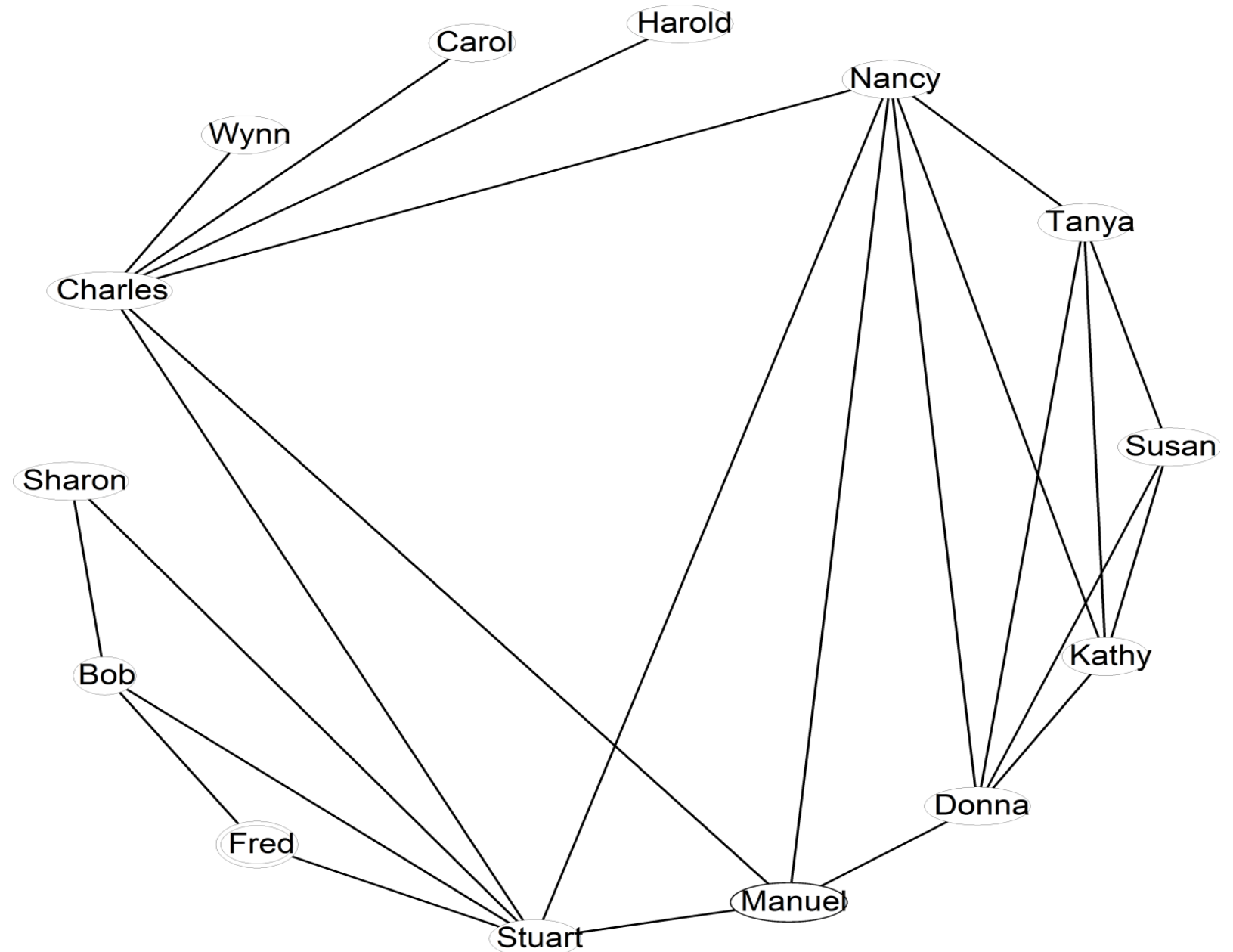
But When It Comes to Trust...



Source: Krackhardt and Hanson, 1993

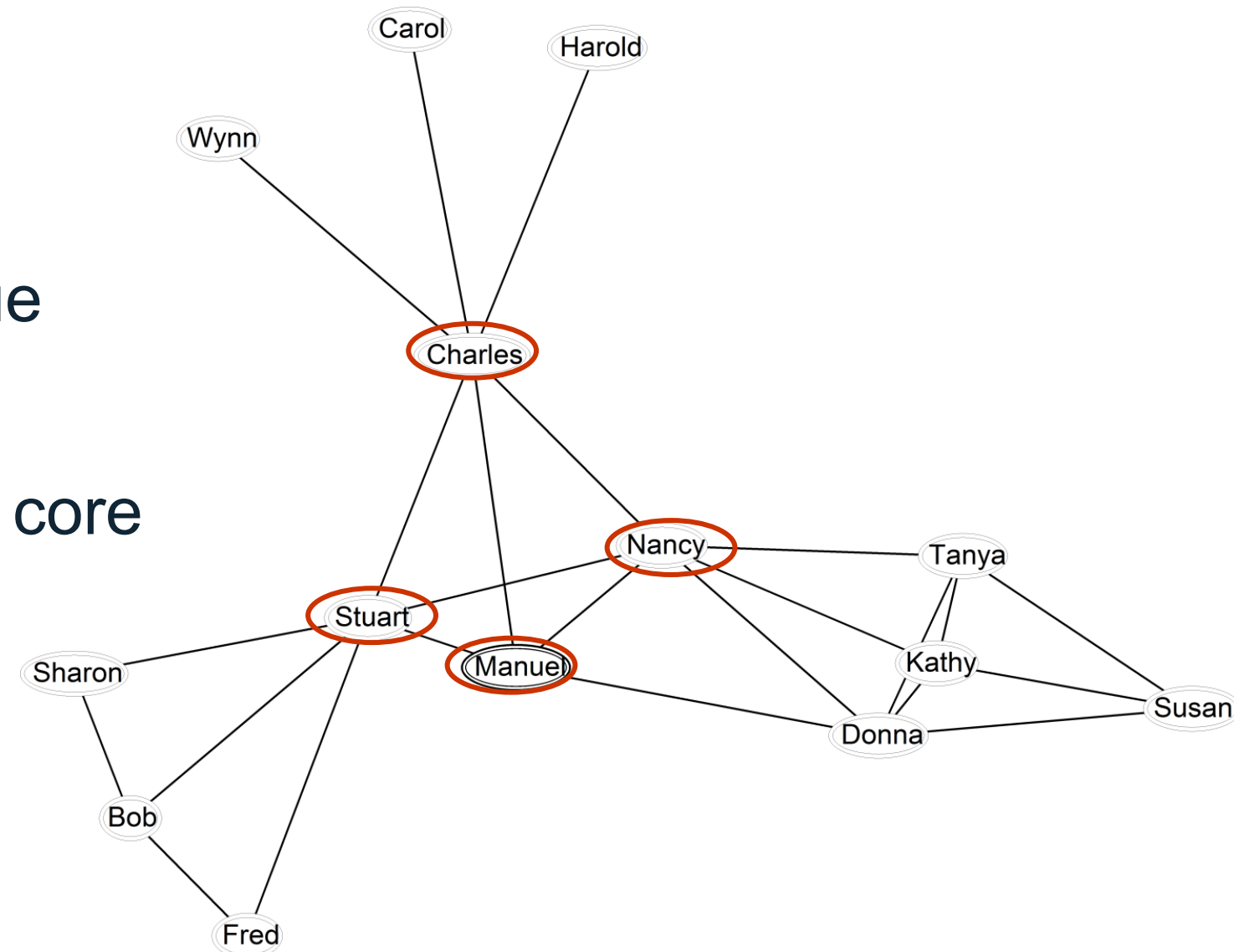
Value of Network Analysis

- A number of network analysis techniques have been developed.
- The aim is to find important characteristics of a network.
- Who in this network is important?
- **Note** nodes can be business partners, such as suppliers
- How well connected is the network
- Is there seamless information flow?



A Simple Visual Modification...

- A simple visual modification to the network can present the important “nodes”
- Analysis can be done in several ways:
 - One could count the links from each node
 - One could also look at nodes which are unique connectors to outliers
 - One could look at nodes which form a central core cluster
 - One could look at the strength of each link



Some Metrics of Interest: Degree Centrality

- Represents the number of connections a node has:

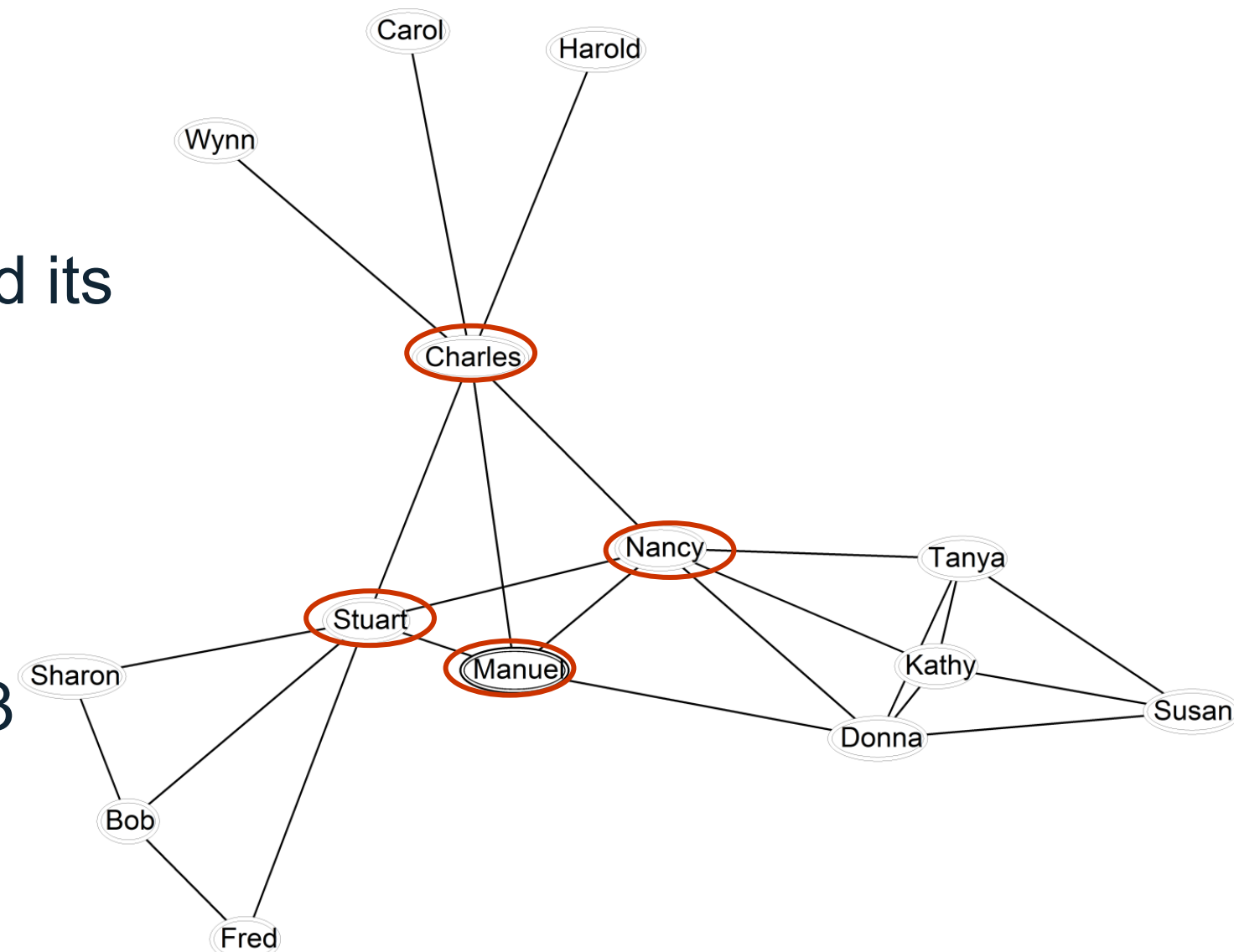
- For example, Wynn has a degree of 1 and Charles has a degree of 6

- Degree represents the centrality of a node and its importance to the network

- They can be important to information flow

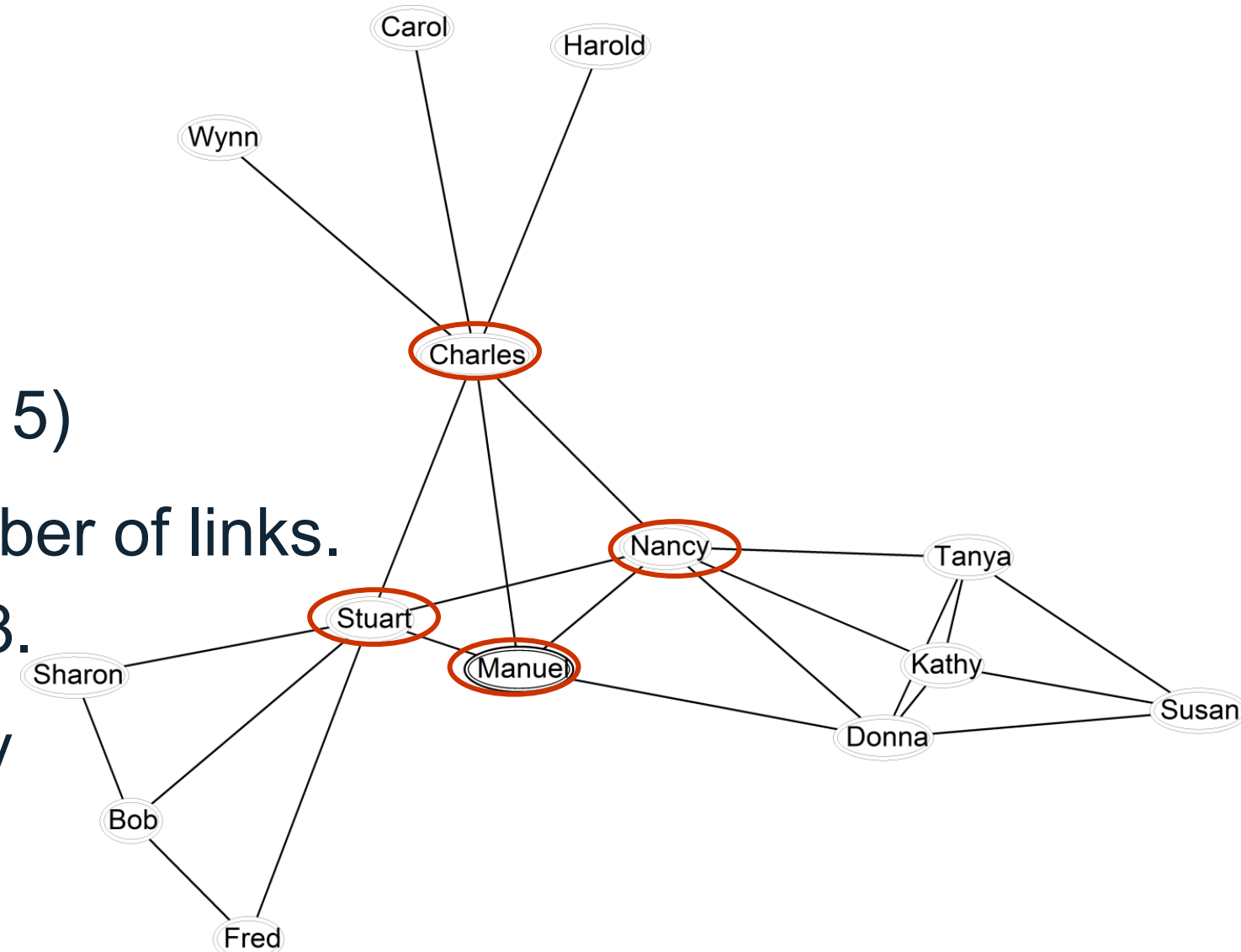
- The average network degree:

$$1+1+1+6+6+2+3+2+4+6+4+4+5+3/14=3.43$$
shows how interconnected the network is



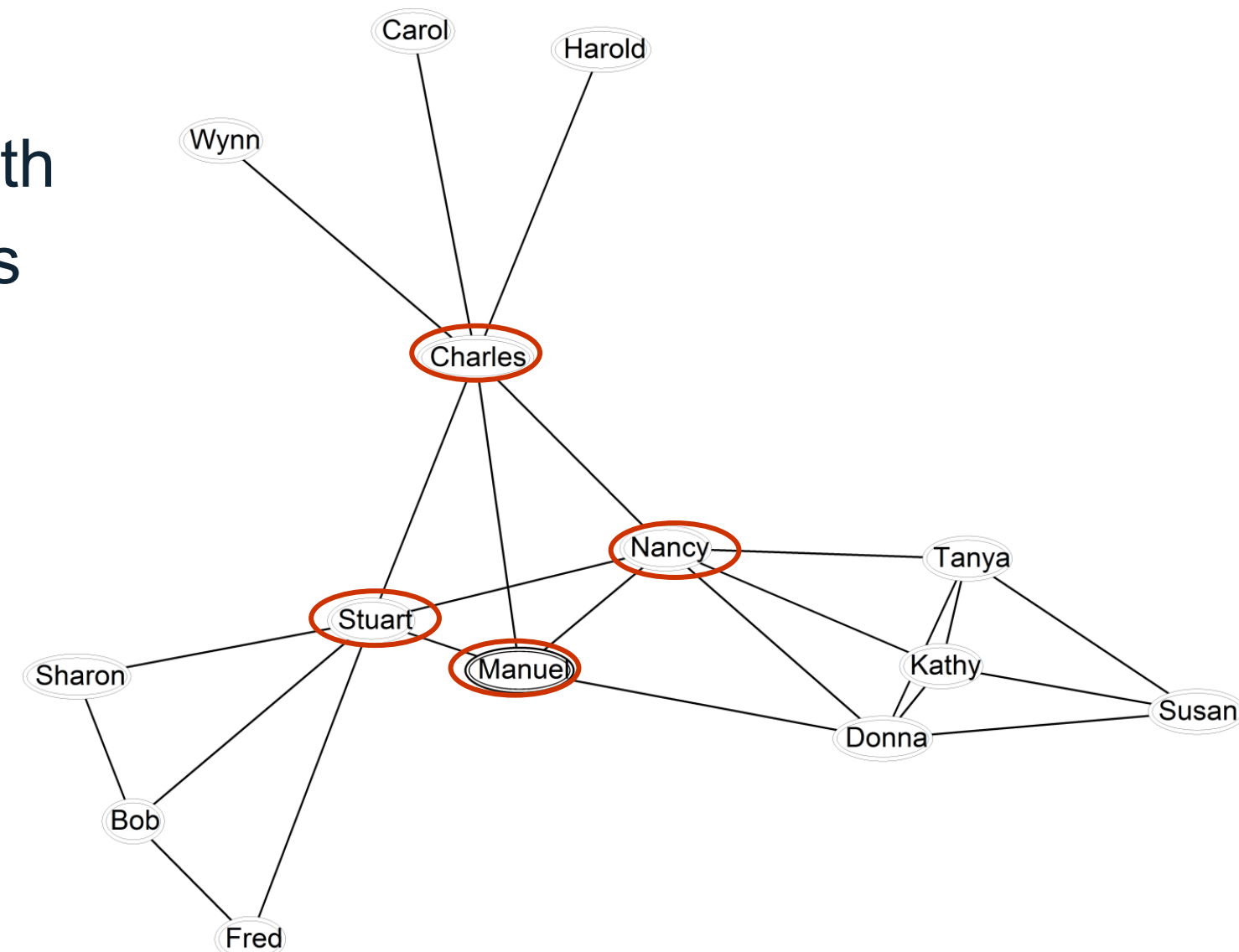
Metrics of Interest: Paths and Shortest Path

- Any two nodes in a network connect to each other through paths
- Wynn is connected to Donna through
 - Charles-Nancy (path length = 3)
 - Charles-Stuart-Manuel (path length = 4)
 - Charles-Nancy-Tanya-Kathy (path length = 5)
- The shortest path is the path with fewest number of links.
In this case: Wynn-Charles-Nancy-Donna or 3.
- The shortest path is a measure of how quickly information may travel through the network



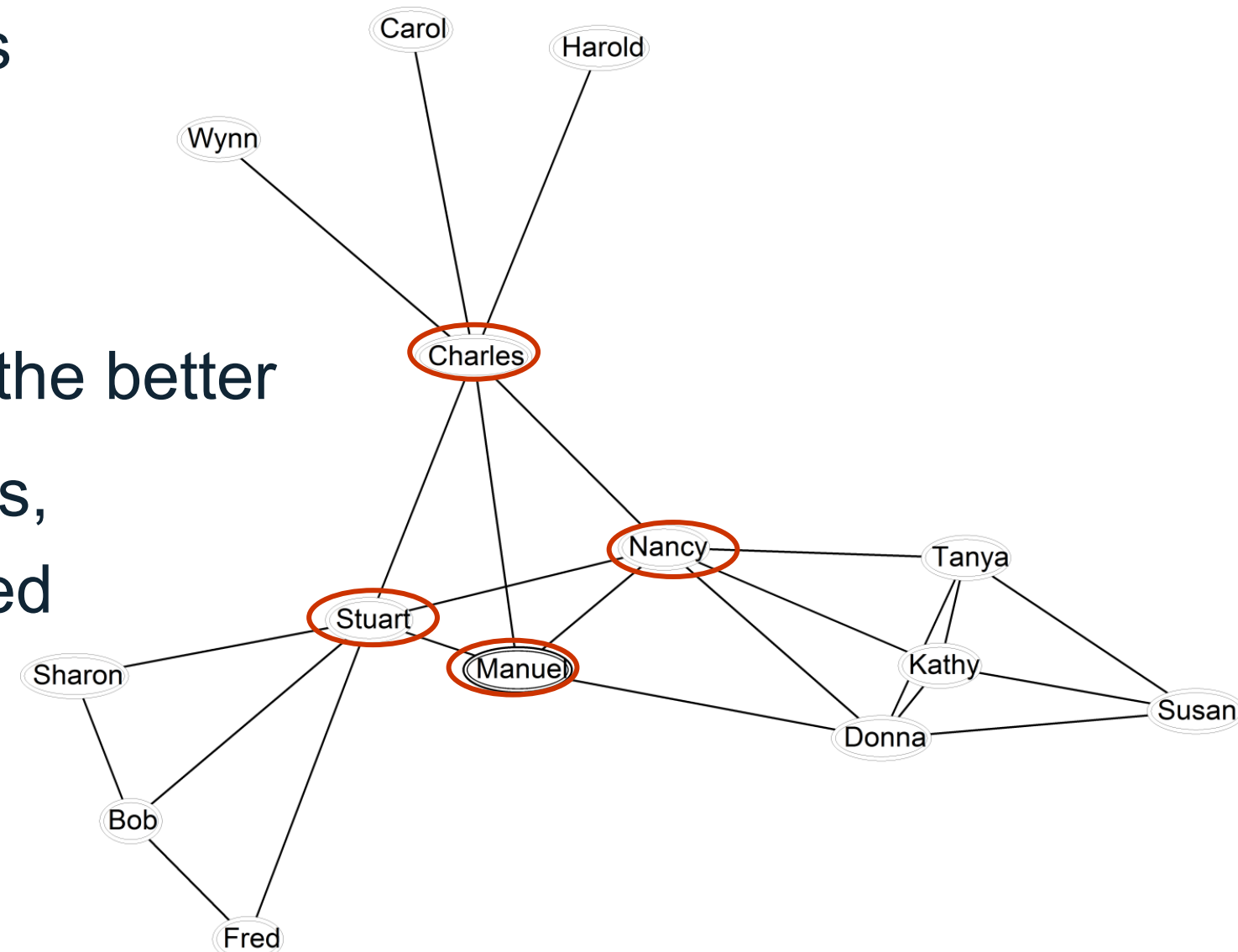
Metrics of Interest: Between-ness Centrality

- Betweenness measures how many Shortest Paths pass through a node
- For example: all Shortest paths associated with Wynn, Carol and Harold pass through Charles
- Charles has a betweenness > 3
- Wynn has a betweenness $= 0$
- Betweenness is a measure of how important a node is to connectivity in a network



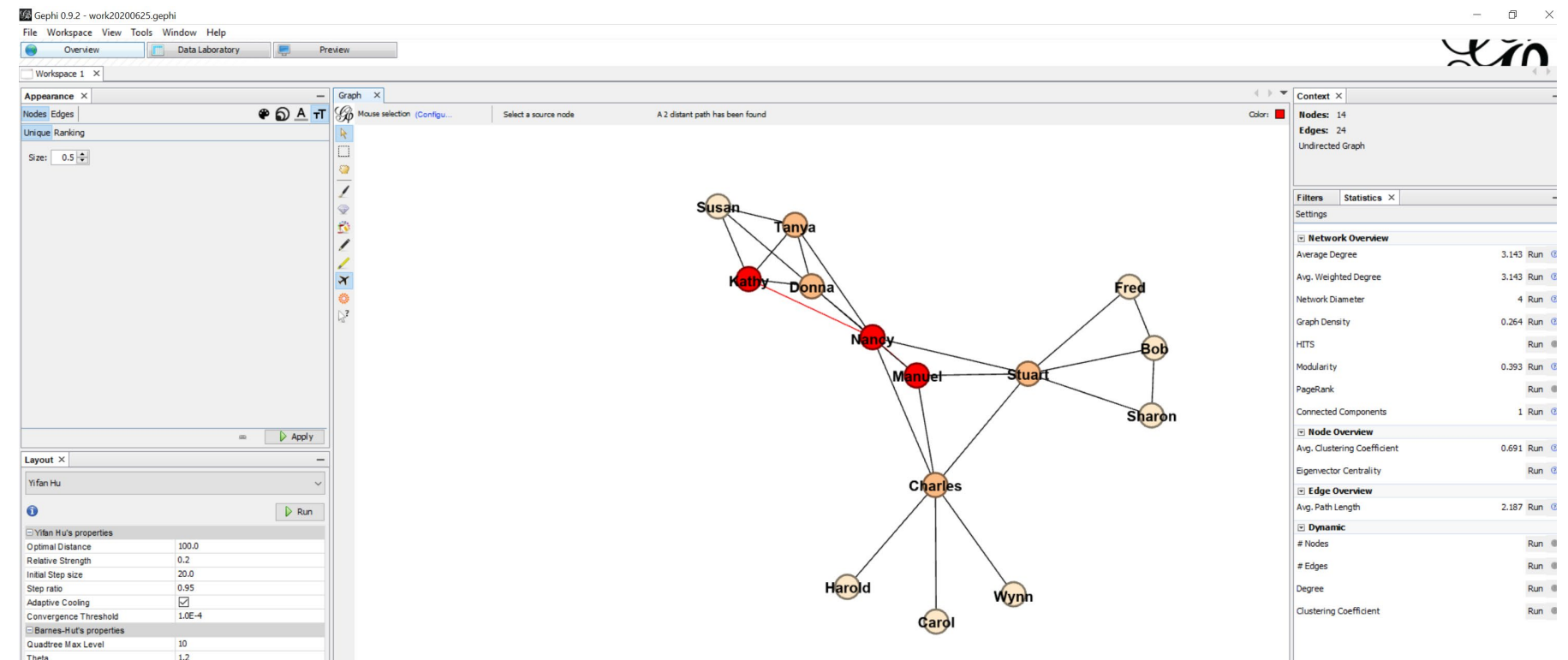
Metrics of Interest: Eigenvector Centrality

- While closeness and degree show how connected a node is, they do not represent how important a node is
- Importance comes from how close one is to important nodes
- The closer one is to highly connected nodes, the better
- For example Donna is connected to five nodes, but most of these nodes are not well connected
- However, Manuel is connected to four nodes, but three of them are highly connected



Analysis Using Gephi

- We can use an open source program to analyse metrics of importance to social networks
- Gephi is quite simple to learn (see video) and I used it to analyse our example
- There are examples of Gephi usage to analyse one's Facebook Social graph search for Gephi on Youtube

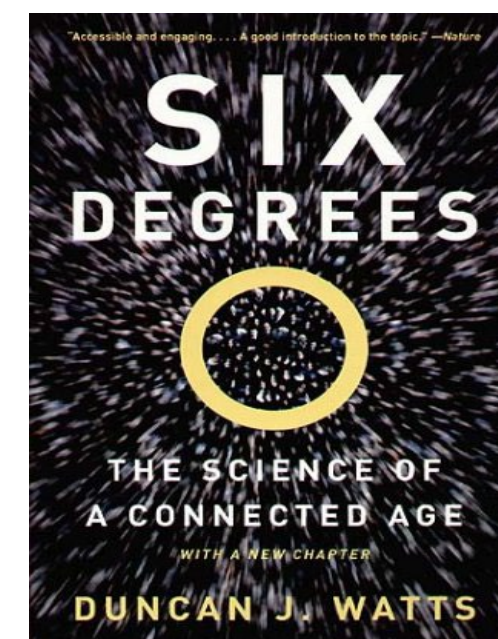
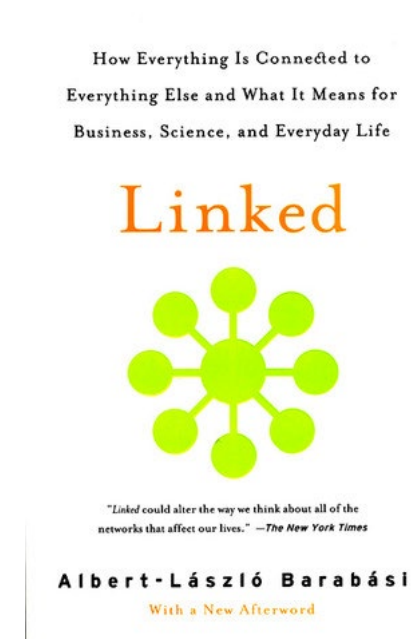


Network Information

- On an individual basis:
 - “You are who your friends are”: A knowledge of one’s connection can be a valuable source of information to a business
- On a local basis
 - A knowledge of a group of connections can be of interest to marketers and advertisers, etc.
- On a global basis
 - Trends discovery can be of interest to a large group of people. Politicians, companies, Hollywood, Military, etc.

Social Networks as Popular Science

- Social Networks is a new scientific field with applications in a diverse area, from sociology to epidemiology to marketing etc.
- Theories have been developed to analyze networks and used to find out how networks behave



Applications to Team Design

- Understanding group dynamics is useful in designing good teams
- Examples in sport are many: While “Moneyball” like projects are mostly individual teams, some sports are more team-oriented, e.g. basketball, football etc
- In these team sports who plays alongside who in which position matters

<i>Social Actor</i>	<i>Strengths</i>	<i>Weaknesses</i>	<i>Personal Motives</i>	<i>Personal Situation</i>
Faizan	Critical thinker	Overthinks things	Monetary incentives	?
Pedro	Good with people	Usually unfocused	Increased social status	?
Rahina	Numbers expert	Not a public speaker	Monetary incentives	Mortgage loan
Vera	Sees “the big picture”	Poor attention to detail	Wants to leave a legacy	Retiring soon

- In designing work teams, members’ capabilities and strengths/weaknesses etc need to be taken into account

Source: [Monzani 2020](#)

How the Members Work Together?

- The next step is whether members work well together towards a needed goal

	John	Paul	Jane	Mary
Faizan	–	Pedro – > Faizan	<i>Rahina</i> < – > <i>Faizan</i>	Vera – ? – Faizan
Pedro	Faizan –> Pedro	–	Rahina – ? – Pedro	Vera – ? – Pedro
Rahina	<i>Faizan</i> < – > <i>Rahina</i>	Pedro – > Rahina	–	Vera – ? – Rahina
Vera	Faizan – ? – Vera	<i>Pedro</i> –> <i>Vera</i>	<i>Rahina</i> –> <i>Vera</i>	–

- An organisation develops over time a sense of who works well with who and which teams deliver results
- Professional firms (Deloitte, Accenture, etc) put teams of people of together for different tasks, and dismantle/re-form teams on regular basis.

Summary

- Humans have managed to create sophisticated, hierarchical societies, helping them build magnificent structures and technologies
- These social inter-relations helps us work together successfully
 - A bad social order can lead to failures
- The field of Science of Networks helps us understand work dynamics and how information flows through organisations and alliances